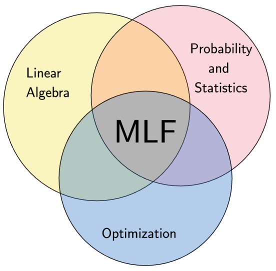
# Week-5

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## 1. Complex Numbers

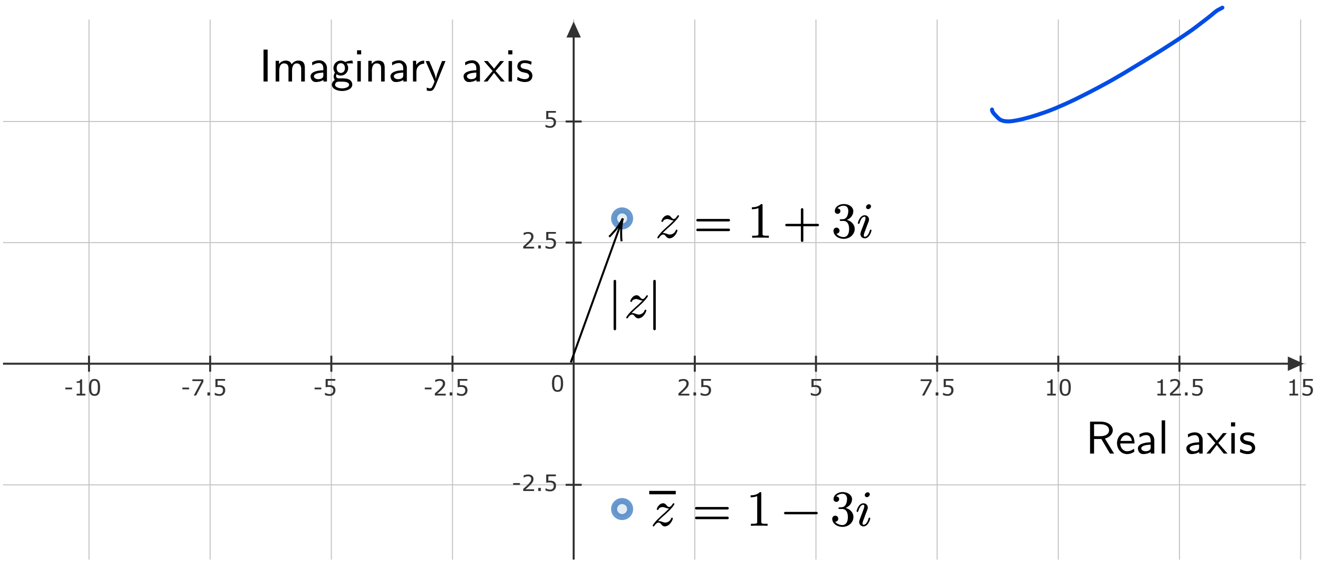
Define $i = \sqrt{-1}$. So $i^2 = -1$. A complex number is of the form $x + iy$, where $x, y \in \mathbb{R}$. For example, the following are all complex numbers:

- $1 + i$
- $3 + 5i$
- $1 - \sqrt{2}i$
- $3$
- $0$

The set of all complex numbers is denoted by $\mathbb{C}$. Real numbers are a subset of complex numbers: $\mathbb{R} \subset \mathbb{C}$. Complex numbers can be plotted on the 2D plane, often called the Argand plane.

Some operations on complex numbers:

- Conjugate: If $z = x + iy$ is a complex number, then $\overline{z} = x - iy$. Geometrically, $\overline{z}$ is the reflection of $z$ about the x-axis (real axis).
  - For example, if $z = 2 + 5i$, then its conjugate is $\overline{z} = 2 - 5i$.
  - For example, if $z = i$, then $\overline{z} = -i$
  - For a real number, $z = \overline{z}$.




- Modulus or the absolute value: If $z = x + iy$, then $|z| = \sqrt{x^2 + y^2}$, and this is the distance of the complex number $z$ from the origin.

A few important points:

- For every complex number $z$, $|z|^2 = z\overline{z}$.
- $|z|$ is a non-negative real number and is zero if and only if $z = 0$
- $z$ is a real number if and only if $z = \overline{z}$

## 2. Complex Vector Spaces

$\mathbb{C}^n$ is the complex equivalent of $\mathbb{R}^n$. For example, let us look at $\mathbb{C}^2$

$$\mathbb{C}^2 = \{(a, b) : a, b \in \mathbb{C}\}$$

For example:

- $(1 + i, 1 - i) \in \mathbb{C}^2$
- $(3, 4) \in \mathbb{C}^2$
- $(5, 3i) \in \mathbb{C}^2$

Vector addition in $\mathbb{C}^2$:

$$(1 + i, 2 - 3i) + (i, -3i) = (1 + 2i, 2 - 6i)$$

Scalar multiplication in $\mathbb{C}^2$:

$$(3i) \cdot (1+i, 3-i)$$

$$(1-i) \cdot (1+i, 3-i)$$

$$3 \cdot (1+i, 3-i)$$

> **Remark:** In a vector space, the scalars come from a set called a **field**. For real vector spaces, the field is $\mathbb{R}$. For example, in $\mathbb{R}^3$, the field is $\mathbb{R}$. In complex vector spaces, the "field" is $\mathbb{C}$. So for $\mathbb{C}^3$, the field is $\mathbb{C}$, that is, the scalars are going to come from $\mathbb{C}$.

In complex vector space, without additional structure, you can only perform two operations:

- addition
- scalar multiplication

If you want to discuss geometric ideas such as length and angles, you need to introduce a new operation, the inner product. For real vector spaces, the dot product was one such inner product, $x, y \in \mathbb{R}^2$, then $x^T y$ was the dot-product. The dot-product will not work for complex vector spaces. To see why, recall that $x^T x$ is the squared length of a vector. For a complex vector, let us see what $x^T x$ gives. If $x = (i, i)$, then:

$$x^T x = \begin{bmatrix} i & i \end{bmatrix} \begin{bmatrix} i \\ i \end{bmatrix} = 2i^2 = -2$$

The squared length of a complex number turns out to be negative. So the simple dot product is not adequate to discuss geometric ideas in complex vector spaces.

## 3. Complex Dot Product

In a complex vector space, a valid inner product is given as follows. If $x, y \in \mathbb{C}^n$, then the complex dot product is defined as:

$$\overline{x}^T y$$

$\overline{x}^T$ is the conjugate transpose of $x$ $\rightarrow$ we first perform the conjugate and then take the transpose. But these operations could also be interchanged so that $\overline{x}^T = \overline{x^T}$.

Let us take some examples in $\mathbb{C}^2$.

$$x = \begin{bmatrix} 1+i \\ 1-i \end{bmatrix}, \ y = \begin{bmatrix} 2i \\ 3-i \end{bmatrix}$$

$$\begin{aligned}
\overline{x}^T y &= \overline{\begin{bmatrix} 1+i \\ 1-i \end{bmatrix}}^T \begin{bmatrix} 2i \\ 3-i \end{bmatrix} \\
&= \overline{\begin{bmatrix} 1+i & 1-i \end{bmatrix}} \begin{bmatrix} 2i \\ 3-i \end{bmatrix} \\
&= \begin{bmatrix} \overline{1+i} & \overline{1-i} \end{bmatrix} \begin{bmatrix} 2i \\ 3-i \end{bmatrix} \\
&= \begin{bmatrix} 1-i & 1+i \end{bmatrix} \begin{bmatrix} 2i \\ 3-i \end{bmatrix} \\
&= (1-i)2i + (1+i)(3-i) \\
&= 2i - 2i^2 + 3 + 2i - i^2 \\
&= 6 + 4i
\end{aligned}$$

**Notation**

It is a bit cumbersome to keep writing $\overline{x}^T$. So a simpler notation is $x^*$.

$$x^* := \overline{x}^T$$

In $x^* y$, there are two arguments:

- $x$ is the first
- $y$ is the second

**Properties**

- Linearity in second argument:
  - $x^*(y+z) = x^* y + x^* z$

- $x^*(cy) = c(x^*y)$
- Definiteness
  - $x^*x \geqslant 0$ and $x^*x = 0$ if and only if $x = 0$
  - Intuition: In $\mathbb{R}^2$, $x^Tx = ||x||^2$; $x^Tx$ is capturing the length of the vector $x$. We want something similar in $\mathbb{C}^2$.
- Conjugate symmetry
  - $\overline{x^*y} = y^*x$
  - First thing is, $x^*y \neq y^*x$ in general

Some examples:

$$x = \begin{bmatrix} 1+i \\ 1-i \end{bmatrix}$$

$$x^*x = \begin{bmatrix} 1-i & 1+i \end{bmatrix} \begin{bmatrix} 1+i \\ 1-i \end{bmatrix}$$

$$(1-i)(1+i) + (1+i)(1-i) = 4$$

The length of the vector of $x$ is $2$.

Note:

- $(x^*)^* = x$
  - In real vector spaces, this is the same as doing $(x^T)^T = x$
- $\overline{x^T} = \overline{x}^T$
  - One is changing a row to a column
  - The other is doing a component wise conjugate
- $(x^*y)^* = y^*x$
  - Note that $x^*y$ is a scalar. So transposing it will leave it a scalar. However, taking the conjugate will invoke the third property: conjugate symmetry.
  - The real version of this is $(x^Ty)^T = y^Tx$

A note concerning the product $(\lambda x)^* y$

$$\begin{aligned}(\lambda x)^* y &= (\overline{\lambda x})^T y \\ &= \overline{\lambda}\,\overline{x}^T y \\ &= \overline{\lambda} x^* y\end{aligned}$$

Be careful with respect to taking $\lambda$ out of the first argument. It comes out as $\overline{\lambda}$.

## 4. Complex Matrices

$$A \in \mathbb{C}^{m \times n}$$

An example:

$$A = \begin{bmatrix} 1+i & 1 & 3-4i \\ i & 0 & 5 \end{bmatrix}$$

Just like we had $x^* = \overline{x}^T$ for vectors, we have $A^* = \overline{A}^T$ for matrices. For example, $A^*$:

$$A^T = \begin{bmatrix} 1+i & i \\ 1 & 0 \\ 3-4i & 5 \end{bmatrix}, \; A^* = \begin{bmatrix} 1-i & -i \\ 1 & 0 \\ 3+4i & 5 \end{bmatrix}$$

Note that $A^T$ is not the same as $A^*$.

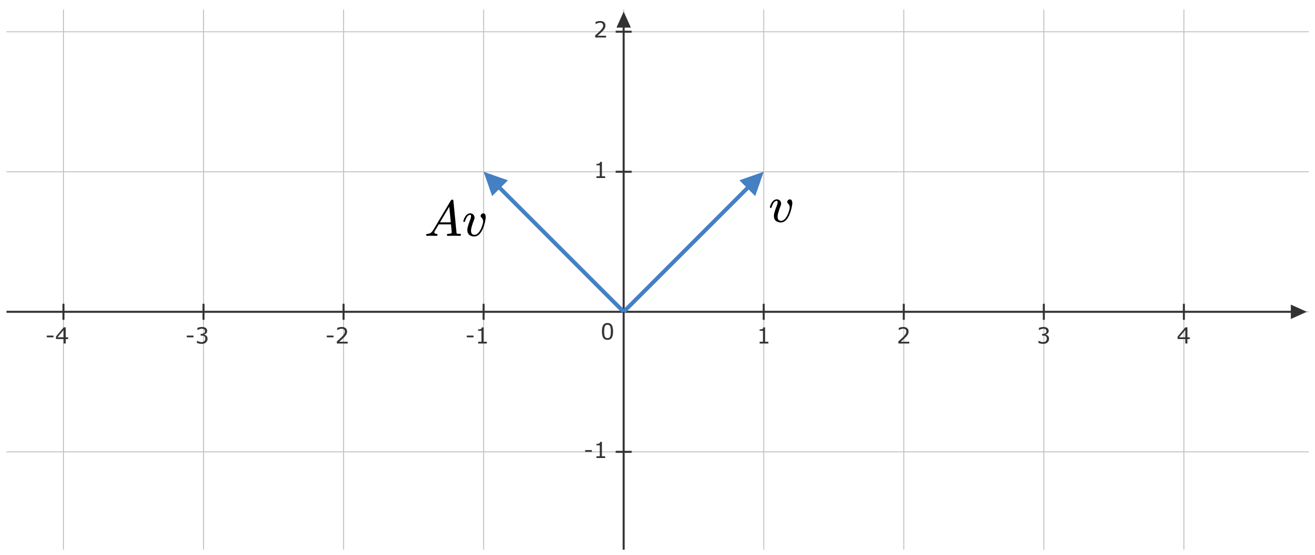
**Theorem:** Every complex square matrix $A \in \mathbb{C}^{n \times n}$ has $n$ eigenvalues, not necessarily distinct.

[To see why this is true, for those interested, refer to the fundamental theorem of algebra, which states that every polynomial of degree $n$ has $n$ complex roots.]

Why is this theorem important? For real matrices, there is no guarantee that every matrix will have $n$ real eigenvalues.

In quiz-1, $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, where $0 < \theta < 180^\circ$, has no real eigenvalues.

What does $A$ do to vectors in $\mathbb{R}^2$? It is a rotation matrix. It rotates vectors anti-clockwise by $\theta$. $v \in \mathbb{R}^2$, then $Av$ is going to be rotated by $\theta$ degrees.




If $v$ has to be an eigenvector of $A$, then $v$ and $Av$ should point in the same direction. This means that the angle between $v$ and $Av$ is either $0^\circ$ or $180^\circ$. Therefore, $A$ cannot have any eigenvectors.

## 5. Hermitian Matrices

A Hermitian matrix is the complex version of a real symmetric matrix. A Hermitian matrix $A$ is a square complex matrix which is equal to its conjugate transpose.

$$A = A^*$$

For example:

$$A = \begin{bmatrix} 1 & 1+i \\ 1-i & 3 \end{bmatrix}$$

$$
\begin{aligned}
A^* &= \begin{bmatrix} 1 & 1+i \\ 1-i & 3 \end{bmatrix} \\
&= A
\end{aligned}
$$

Let us look at all possible Hermitian matrices of shape $2 \times 2$:

$$
A = \begin{bmatrix} a_1 + ia_2 & b_1 + ib_2 \\ c_1 + ic_2 & d_1 + id_2 \end{bmatrix}
$$

$$
\begin{aligned}
A^* &= \begin{bmatrix} a_1 - ia_2 & c_1 - ic_2 \\ b_1 - ib_2 & d_1 - id_2 \end{bmatrix} \\
&= \begin{bmatrix} a_1 + ia_2 & b_1 + ib_2 \\ c_1 + ic_2 & d_1 + id_2 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
a_1 - ia_2 &= a_1 + ia_2 \\
c_1 - ic_2 &= b_1 + ib_2 \\
c_1 + ic_2 &= b_1 - ib_2 \\
d_1 - id_2 &= d_1 + id_2
\end{aligned}
\implies
\begin{aligned}
a_2 &= 0 \\
c_1 &= b_1 \\
c_2 &= -b_2 \\
d_2 &= 0
\end{aligned}
$$

Therefore:

$$
A = \begin{bmatrix} a & b + ic \\ b - ic & d \end{bmatrix}
$$

**Remark:** If $A \in \mathbb{C}^{n \times n}$ is Hermitian, then the diagonal elements of $A$ are real.

**Properties**

For all the properties below, assume that $A$ is Hermitian:

- If all the entries of $A$ are real, then $A$ is a real symmetric matrix.

$$A^* = \overline{A}^T = A^T$$

- All the eigenvalues of $A$ are real. Let $\lambda$ be an eigenvalue of $A$ with eigenvector $v$. [Note: $v \neq 0$]

  **Proof:** Starting with $Av = \lambda v$, we multiply both sides by $v^*$:

$$\begin{aligned}
\lambda v &= Av \\
v^*(\lambda v) &= v^* A v \\
\lambda\big(v^* v\big) &= v^* A v
\end{aligned}$$

  We now make use of $A^* = A$ and the fact that $(Av)^* = v^* A^*$:

$$\begin{aligned}
\lambda\big(v^* v\big) &= v^* A^* v \\
&= (Av)^* v \\
&= (\lambda v)^* v \\
&= \overline{\lambda} v^* v
\end{aligned}$$

  Therefore:

$$\begin{aligned}
\lambda\big(v^* v\big) &= \overline{\lambda}\big(v^* v\big) \\
\big(v^* v\big)(\lambda - \overline{\lambda}) &= 0
\end{aligned}$$

  Can $v^* v = 0$? No, because $v$ is an eigenvector of $A$. This means that $\lambda = \overline{\lambda}$. It follows that $\lambda$ is real. As an aside, if $A$ is real symmetric, then its eigenvalues are real.

**Remark:**
- If $A$ is real symmetric, then $A$ is also Hermitian.
- If $A$ is Hermitian, then $A$ is not necessarily real symmetric.

- Eigenvectors corresponding to distinct eigenvalues are orthogonal.

  **Proof:** If $(\lambda_1, v_1)$ and $(\lambda_2, v_2)$ are two pairs with $\lambda_1 \neq \lambda_2$. We want show that $v_1^* v_2 = 0$. We start with $Av_1 = \lambda_1 v_1$ and multiply both sides with $v_2^*$:

$$\begin{aligned} Av_1 &= \lambda_1 v_1 \\ v_2^* A v_1 &= v_2^*(\lambda_1 v_1) \\ &= \lambda_1 \left(v_2^* v_1\right) \end{aligned}$$

Once again, we make use of $A^* = A$

$$\begin{aligned} \lambda_1 \left(v_2^* v_1\right) &= v_2^* A v_1 \\ &= v_2^* A^* v_1 \\ &= (Av_2)^* v_1 \\ &= (\lambda_2 v_2)^* v_1 \\ &= \overline{\lambda_2} v_2^* v_1 \\ &= \lambda_2 \left(v_2^* v_1\right) \end{aligned}$$

Finally:

$$(\lambda_1 - \lambda_2)\left(v_2^* v_1\right) = 0$$

It follows that $v_2^* v_1 = 0$ and hence $v_1 \perp v_2$.

## 6. Unitary Matrices

A unitary matrix $U \in \mathbb{C}^{n \times n}$ is one that has orthonormal columns. For example:

$$U = \frac{1}{5}\begin{bmatrix} 3 & 4i \\ 4i & 3 \end{bmatrix}$$

The columns of $U$ are:

Orthonormal = Unit length + Orthogc

- $u_1 = \frac{1}{5}(3, 4i)$
  - $u_1^* u_1 = \dfrac{3 \times 3 + (4i)(-4i)}{25} = 1$

- $u_1$ has unit norm

- $u_2 = \frac{1}{5}(4i, 3)$
  - $u_2^* u_2 = \dfrac{(-4i)(4i) + 3 \times 3}{25} = 1$
  - $u_2$ has unit norm

- $u_1 \perp u_2$

$$u_2^* u_1 = \frac{(-4i) \times 3 + 3 \times (4i)}{25} = 0$$

If $U$ is unitary, then $U^*U = I$. Let us show this for the case of $2 \times 2$ unitary matrices. If $U$ can be written as follows:

$$U = \begin{bmatrix} | & | \\ u_1 & u_2 \\ | & | \end{bmatrix} \Longrightarrow U^* = \begin{bmatrix} - & u_1^* & - \\ - & u_2^* & - \end{bmatrix}$$

Then:

$$\begin{aligned} U^*U &= \begin{bmatrix} - & u_1^* & - \\ - & u_2^* & - \end{bmatrix} \begin{bmatrix} | & | \\ u_1 & u_2 \\ | & | \end{bmatrix} \\ &= \begin{bmatrix} u_1^* u_1 & u_1^* u_2 \\ u_2^* u_1 & u_2^* u_2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= I \end{aligned}$$

It follows that $UU^* = I$ and $U^{-1} = U^*$. In fact, a unitary matrix can be defined in any one of the following ways, all of which are equivalent:

- $U$ has orthonormal columns
- $U^*U = I$
- $UU^* = I$

- $U^{-1} = U^*$

The real counterpart of a unitary matrix is an orthogonal matrix. If all the entries of a unitary matrix are real, then $U^* = U^T$. We usually denote orthogonal matrices by $Q$. An orthogonal matrix has the following equivalent definitions:

- $Q$ has orthonormal columns
- $Q^T Q = I$
- $QQ^T = I$
- $Q^{-1} = Q^T$

**Properties**

- Unitary matrices preserve the inner product:

  **Proof:** Let us start with $x, y \in \mathbb{C}^n$. We need to show that $(Ux)^*(Uy) = x^*y$

$$\begin{aligned}(Ux)^*(Uy) &= x^*U^*Uy \\ &= x^*Iy \\ &= x^*y\end{aligned}$$

  From this, we can see that $||Ux||^2 = ||x||^2$. Unitary matrices preserve lengths.

- The eigenvalues of a unitary matrix have absolute value 1.

  **Proof:** Let $(\lambda, v)$ be a pair of eigenvalue-eigenvector, then we can show that $\left(\frac{1}{\lambda}, v\right)$ is an eigenvalue-eigenvector pair for $U^*$:

$$\begin{gathered}Uv = \lambda v \\ U^*Uv = \lambda U^*v \\ \boxed{U^*v = \frac{1}{\lambda}v}\end{gathered}$$

Now consider:

$$
\begin{aligned}
(Uv)^* &= (\lambda v)^* \\
\boxed{v^* U^* = \overline{\lambda} v^*}
\end{aligned}
$$

Multiplying both sides by $\lambda v$ on the right and using the first result:

$v^*v = |\lambda|^2 v^*v$

$$
\begin{aligned}
v^* U^* (\lambda v) &= \overline{\lambda} v^* (\lambda v) \\
\frac{v^* v}{\lambda} &= \overline{\lambda} v^* v \\
v^* v (\lambda \overline{\lambda} - 1) &= 0 \\
v^* v \big(|\lambda|^2 - 1\big) &= 0
\end{aligned}
\implies |\lambda| = 1
$$

And that completes the proof.

Let us verify the second property with an example. If $U$ is:

$$
U = \frac{1}{5}\begin{bmatrix} 3 & 4i \\ 4i & 3 \end{bmatrix}
$$

The characteristic polynomial of $U$ is:

$$
\begin{aligned}
|U - \lambda I| &= \begin{vmatrix} \frac{3}{5} - \lambda & \frac{4i}{5} \\ \frac{4i}{5} & \frac{3}{5} - \lambda \end{vmatrix} \\
&= \left(\frac{3}{5} - \lambda\right)^2 + \frac{16}{25}
\end{aligned}
$$

The eigenvalues of $U$ are:

$$
\boxed{\frac{3}{5} - \lambda = \pm \frac{4}{5} i}
$$

$$
\boxed{\lambda = \frac{3 + 4i}{5}, \frac{3 - 4i}{5}}
$$

Note that $\lambda$ is complex, but $|\lambda| = 1$, as expected.

## 7. Spectral Theorem

### Complex version

**Spectral Theorem**

If $A$ is a Hermitian matrix, then there exists a diagonal matrix $D$ and a unitary matrix $U$ such that $A = UDU^*$. That is, every Hermitian matrix can be unitarily diagonalized.

- Every Hermitian matrix can be diagonalized with the help of a unitary matrix. $A = UDU^*$ is called the spectral decomposition of $A$.
- The diagonal entries of $D$ contain the real eigenvalues of $A$ and the columns of $U$ contain the corresponding eigenvectors of $A$.

### Real version

If $A$ is a symmetric matrix, then there exists a diagonal matrix $D$ and an orthogonal matrix $Q$ such that $A = QDQ^T$. That is, every real symmetric matrix can be orthogonally diagonalized.

Alternatively, every symmetric matrix can be diagonalized with the help of an orthogonal matrix. $A = QDQ^T$ is called the spectral decomposition of $A$.

Example for the complex case

$$A = \begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix}$$

**Step-1: Eigenvalues**

$$
\begin{aligned}
|A-\lambda I| &= \begin{vmatrix} 1-\lambda & 1+i \\ 1-i & 2-\lambda \end{vmatrix} \\
&= (1-\lambda)(2-\lambda)-(1+i)(1-i) \\
&= (1-\lambda)(2-\lambda)-2 \\
&= \lambda^2-3\lambda
\end{aligned}
$$

Therefore, $\lambda = 3, 0$.

**Step-2: Eigenvectors**

**Sub-step-1:** $\lambda = 0$

$$\begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$R_2 \to R_2 - (1-i)R_1$$

$$\begin{bmatrix} 1 & 1+i \\ 0 & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Note:

$$2-(1-i)(1+i) = 2-\left(1-i^2\right) = 0$$

Therefore:

$$v_1 + (1+i)v_2 = 0$$

$v_2 = -1$, then $v_1 = 1+i$. Therefore, $(1+i, -1)$ is an eigenvector for $\lambda = 0$.

**Sub-step-2:** $\lambda = 3$

$$\begin{bmatrix} -2 & 1+i \\ 1-i & -1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$R_2 \to 2R_2 + (1-i)R_1$$

$$2 - 2i - 2(1-i) = 0$$

$$-2 + (1-i)(1+i) = 0$$

One equation:

$$-2v_1 + (1+i)v_2 = 0$$

$v_2 = 2, v_1 = 1 + i$, therefore, $(1+i, 2)$ is an eigenvector of $A$ corresponding to $\lambda = 3$.

**Step-3: Decomposition**

$$P = \begin{bmatrix} 1+i & 1+i \\ -1 & 2 \end{bmatrix}, \; D = \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix}$$

The following is just an eigen-decomposition ~~which exists for~~ any diagonalizable matrix.

$$\boxed{A = PDP^{-1}}$$

But what we are guaranteed is a spectral decomposition.

**Step-4: Spectral Decomposition**

To get a spectral decomposition, we need to a unitary matrix.

Verify orthogonality:

$$u_1 = (1+i, -1), u_2 = (1+i, 2)$$

$$\begin{aligned} u_1^* u_2 &= (1-i)(1+i) - 2 \\ &= 0 \end{aligned}$$

To get $U$, you have to normalize the columns:

$$u_1 \to \frac{u_1}{||u_1||},\ u_2 \to \frac{u_2}{||u_2||}$$

$$u_1 = (1+i,-1), u_2 = (1+i,2)$$

$$||u_1|| = \sqrt{3}$$
$$||u_2|| = \sqrt{6}$$

$$U = \begin{bmatrix} \frac{1+i}{\sqrt{3}} & \frac{1+i}{\sqrt{6}} \\ \frac{-1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \end{bmatrix},\ U^* = \begin{bmatrix} \frac{1-i}{\sqrt{3}} & \frac{-1}{\sqrt{3}} \\ \frac{1-i}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix}$$

$$\boxed{A = UDU^*}$$

Finally, we have the spectral decomposition:

$$\boxed{\begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix} = \begin{bmatrix} \frac{1+i}{\sqrt{3}} & \frac{1+i}{\sqrt{6}} \\ \frac{-1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{1-i}{\sqrt{3}} & \frac{-1}{\sqrt{3}} \\ \frac{1-i}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix}}$$

- The columns of $U$ are the normalized eigenvectors of $A$
- The diagonal elements of $D$ are the corresponding eigenvalues, which are real in this case.

## 8. Diagonalizability

A summary of what we know about diagonalizability so far:

- All matrices are not diagonalizable.
- All diagonalizable matrices are not unitarily diagonalizable.

- [out-of syllabus] Normal matrices are the class of all unitarily diagonalizable matrices.
- [out-of syllabus] Hermitian matrices and unitary matrices are a subset of normal matrices.

Refer to the following note for more information on normal matrices:

**Out of syllabus, Extra information**

**Normal matrix**: $A$ complex square matrix $A$ is called normal if the following is satisfied:

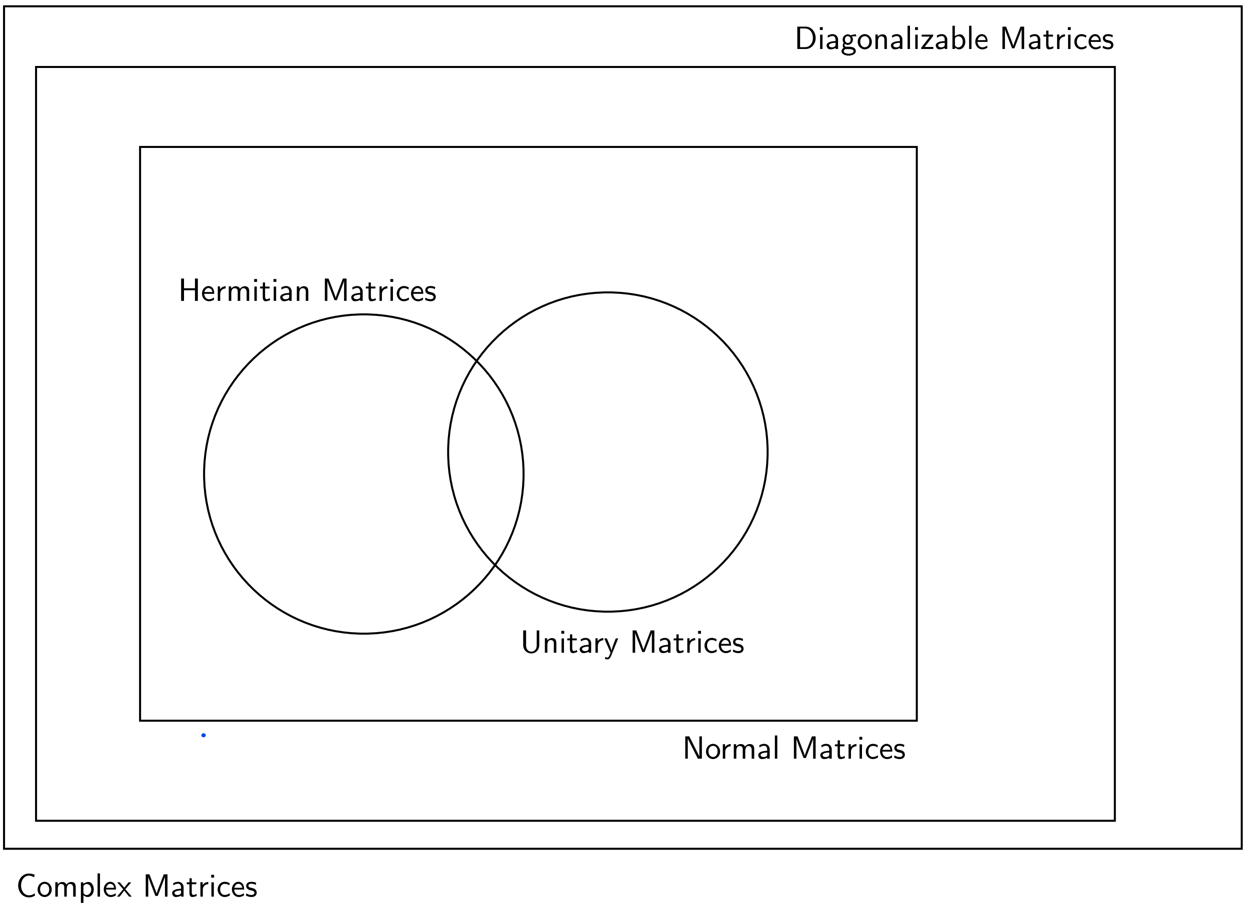
$$\boxed{A^*A = AA^*}$$

- Every Hermitian matrix is normal
- Every Unitary matrix is normal.

**Spectral Theorem:** The (complete) spectral theorem states that every normal matrix is unitarily diagonalizable. In fact, the set of all unitarily diagonalizable matrices is exactly the same as the set of all normal matrices.

The following diagram is from one of these two textbooks:

- Introduction to Linear and Matrix Algebra, Nathaniel Johnston
- Advanced Linear and Matrix Algebra, Nathaniel Johnston




A matrix that is both Hermitian and unitary: $I$

- $I^* = I$
- $I^*I = II^* = I$

Real version. $A$ below is both symmetric and orthogonal:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

## 9. Complex vs Real: a comparison

| Real | Complex | Object |
|---|---|---|
| $\mathbb{R}^n$ | $\mathbb{C}^n$ | Vector space |
| $x^Ty$ (dot product) | $x^*y$ or $\overline{x}^Ty$ (complex dot product) | Inner product |

| | | |
|---|---|---|
| $\sqrt{x^T x}$ | $\sqrt{x^* x}$ | Length<br>---<br>(non-negative) |
| $A^T = A$ | $A^* = A$ | Symmetric vs Hermitian<br>---<br>(real eigenvalues) |
| $Q^T Q = QQ^T = I$<br>$Q^{-1} = Q^T$<br>$\lambda^2 = 1$ | $U^* U = UU^* = I$<br>$U^{-1} = U^*$<br>$\|\lambda\|^2 = 1$ | Unitary vs Orthogonal<br>---<br>(orthonormal columns) |
| $A = QDQ^T$ where $A^T = A$ | $A = UDU^*$ where $A^* = A$ | Spectral decomposition<br>--<br>($D$ is real) |

# 10. Quadratic forms

### Quadratic forms

Symmetric matrices can be used to define what are called quadratic forms.
$f(x) = x^T A x$ is called a quadratic form.

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}$$

$$x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= x_1^2 + 6x_1x_2 + 2x_2^2$$

A few more examples:

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & -2 \\ -1 & -2 & 3 \end{bmatrix}$$

We define the quadratic form $x^T A x$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

$$f(x) = x^T A x = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & -2 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$f(x) = x_1^2 + 2x_2^2 + 3x_3^2 + 6x_1x_2 - 2x_1x_3 - 4x_2x_3$$

To understand how we got here:

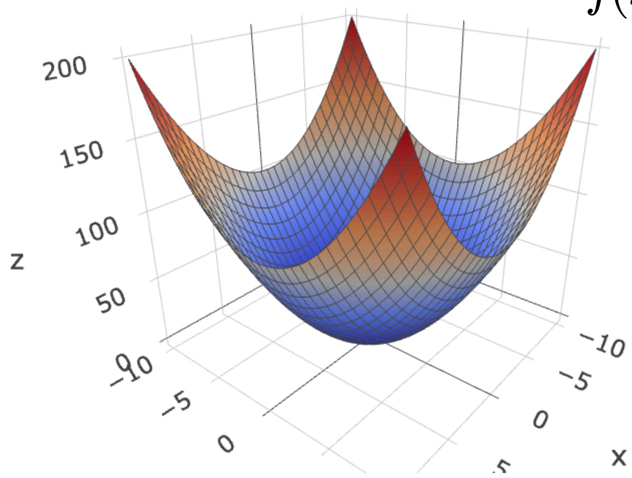
$$\begin{array}{cc} & \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} & \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & -2 \\ -1 & -2 & 3 \end{bmatrix} \end{array}$$

$f$ is a quadratic form in three variables.

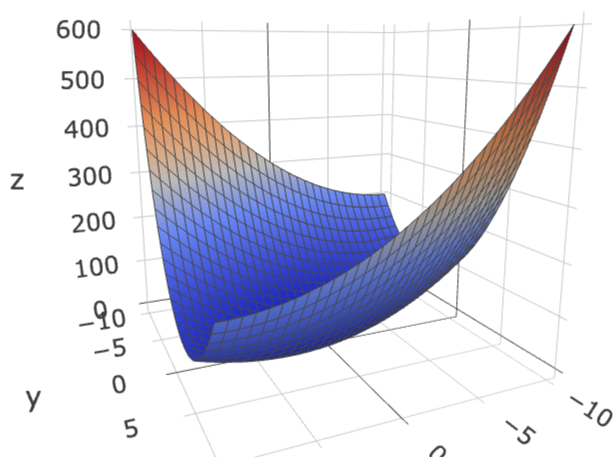
- The coefficients of $x_i^2$ are the diagonal entries of $A$
- The coefficients of $x_i x_j$ are the sum of the corresponding off-diagonal elements, namely, $a_{ij} + a_{ji}$

What do quadratic forms represent?

$$f(x_1, x_2) = x_1^2 + x_2^2$$




$$f(x_1, x_2) = x_1^2 - 2x_1x_2 + 3x_2^2$$




Summary: Every quadratic form can be represented as $f(x) = x^T Ax$, where $A$ is a symmetric matrix. The symmetric matrix $A$ is unique for a given quadratic form.

If you want $f(x)$ to look like a bowl, what properties should it satisfy?

- $f(0) = 0$, meaning, the bowl is touching the XY plane
- No portion of the bowl should go below the XY plane

$$f(x) \geqslant 0$$

This means

$$x^T Ax \geqslant 0$$

This is tied to minimization problems and that is one of the reasons why we are interested in quadratic forms.

# 11. Positive definiteness

A symmetric matrix $A$ is positive definite if $x^T Ax > 0$ for all $x \neq 0$. The corresponding quadratic form (surface) is a bowl shaped surface that touches the "ground" at only the point $x = 0$.

### Properties

We won't prove these completely. Only an outline will be provided. Refer to the notes. A detailed proof will be updated here soon.

- $A$ is positive definite if and only if the eigenvalues of $A$ are positive.

  **Proof outline:** If $(\lambda, v)$ is an eigenpair:

  $$\begin{aligned} Av &= \lambda v \\ v^T Av &= \lambda v^T v \end{aligned}$$

  Since $v^T Av > 0$ for all $v \neq 0$ and since $v^T v > 0$, we see that $\lambda > 0$.

- $A$ is positive definite if and only if $A$ can be expressed as $B^T B$ or $BB^T$ for some matrix $B$.

- **Proof outline**: Since $A$ is symmetric, it is orthogonally diagonalizable.

  $$\begin{aligned} A &= QDQ^T \\ &= \left(QD^{1/2}\right)\left(D^{1/2}Q^T\right) \\ &= BB^T \end{aligned}$$

  Here, $D^{1/2}$ is the matrix whose square is $D$. We can do this because the diagonal elements of $D$ are positive. For example:

  $$D = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$D^{1/2} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \Longrightarrow D = D^{1/2} D^{1/2}$$

- If $A$ is positive definite, then $A$ is invertible.

  **Proof:** Since $A$ has no zero eigenvalues, $A$ is invertible.

**Positive semi-definite**

A symmetric matrix $A$ is positive semi-definite if $x^T Ax \geqslant 0$ for all $x \neq 0$.

$A$ is positive semi-definite if and only if the eigenvalues of $A$ are non-negative.

**Negative definite**

A symmetric matrix $A$ is negative definite if $x^T Ax < 0$ for all $x \neq 0$.

$A$ is negative definite if and only if the eigenvalues of $A$ are negative.

**Negative semi-definite**

A symmetric matrix $A$ is negative Semi definite if $x^T Ax \leqslant 0$ for all $x \neq 0$.

$A$ is negative semi-definite if and only if the eigenvalues of $A$ are non-positive.

**Indefinite matrix**

A symmetric matrix that is neither positive (semi)definite or negative (semi)definite.

A matrix is indefinite if and only if it has at least one positive and at least one negative eigenvalue.